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A developmental psychological perspective on youth trust

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Abstract
Adolescence is a critical developmental period marked by significant social, emotional, and cognitive changes that profoundly impact trust. This paper examines how these developmental processes influence youth trust, as distinct from, and as an antecedent of, adult trust. Adolescence is characterized by significant brain maturation that influences social cognition, emotional regulation, and peer interactions, making it a key period for social trust development. Youth process social information differently than adults, becoming increasingly sensitive to social rewards and peer influence, while parental socialization also shapes youth trust through explicit teachings and implicit behavioral cues. The behaviors and attitudes established during adolescence have lasting impacts on adulthood, such that understanding youth trust has theoretical, practical, and methodological implications for our understanding of trust in adulthood. Current trust theories and methodologies may not fully capture the unique social dynamics of adolescent trust, suggesting the need for more ecologically valid and interdisciplinary research approaches. By integrating insights from neuroscience and psychology, this paper aims to enrich the understanding of youth trust dynamics in youth, underscoring the importance of considering adolescence as a distinct developmental phase in trust research. Ultimately, the paper advocates for a developmental lens in trust research.

Broadly, trust refers to an individual's willingness to accept vulnerability to another person or institution based on the expectation that the trustee will act benevolently, competently, and with integrity [38, 55]. Trust is both a psychological and social construct, which manifests across multiple domains (e.g., interpersonal vs. institutional vs. generalized trust), relational contexts (in-group vs. out-group), and temporal dynamics (initial/one-shot trust vs. trust built through repeated interactions; for a review, see [53]).

Research has made significant strides in understanding the nature and dynamics of trust for adults across contexts [15, 20, 62, 67], but relatively little has considered the implications of developmental science on understanding trust among youth. Although youth are excluded from some aspects of civic life due to their age, youths' trust in individuals, organizations, and institutions has significant implications for their own trust

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behavior and forms the roots of the attitudes and behavior they espouse in adulthood [80].

The nature and process of trust as held by adolescents is necessarily distinct from that of adults. Indeed, classic theoretical perspectives in developmental psychology provide foundations for understanding trust development. Social learning theory [5] posits that individuals acquire behaviors and attitudes through observation, imitation, and modeling of others, particularly influential figures like parents and peers. Erikson's [26] theory of psychosocial development identifies the establishment of basic trust versus mistrust as a fundamental task of early childhood, with implications that extend throughout the lifespan. Attachment theory [9] further elaborates how early relational experiences shape internal working models that guide expectations about others' reliability and responsiveness.

Importantly, adolescence is a time of enormous social, emotional, and cognitive development. During adolescence, structural and functional changes to the adolescent brain underpin psychosocial and cognitive development. As a result, youth process information—particularly social information— differently than adults. These developmental processes have implications for how youth trust others, which is key feature of adolescent social development [18]. Accordingly, developmental research can be applied to create a deeper and more holistic understanding of youth trust, both as a distinct construct and as an antecedent to adult trust. While recent scholarship has attended to the role of neuroscience in adolescent trust development (see review by Krabbendam et al. [48]), this review adds to this conversation by (a) integrating psychosocial and socialization perspectives alongside neuroscience, (b) focusing on how adolescent trust lays the groundwork for adult trust, and (c) offering concrete measurement alternatives and contextual considerations.

1 Adolescent development

Normative neural, biological, and psychosocial changes characterize adolescence the period of the lifespan between childhood and adulthood[1]. Compared to adults, adolescents are less able to consider the consequences of their actions, plan for the future, control their impulses, and regulate their emotions [11]. Additionally, the adolescent brain is highly sensitive to social rewards, easily emotionally aroused, and primed to seek out new sensations [7]. Adolescence is considered the final period of "developmental plasticity," when large-scale transformations in brain circuitry transpire [32].

Maturation takes place across different brain regions on distinct timetables. During adolescence, cognitive development occurs before socioemotional maturation, creating a "gap" in competencies. As a result, adolescents can make rational decisions in unemotional contexts but may struggle to make mature decisions in emotionally-arousing contexts [68]. Indeed, research suggests that adolescents are more likely to act impulsively in the presence of rewarding or threatening stimuli, which may override adult-like behavior controls, relative to neutral conditions [16]. Importantly, a series of studies of 5,000 youth from 11 countries of varied political, cultural, and economic contexts found the

[1] Adolescence is generally considered to comprise ages 10–19. In recognition of the fact that the transition from childhood to adulthood is fluid, the term "youth" is sometimes used in place of "adolescence" to refer to individuals between the ages of 10 and 25, the point at which individuals' brains are considered to have matured to adult levels. When possible, clarify the exact age of participants in cited research, as different researchers may prefer different terms.

same trajectories of sensation-seeking and poor self-control that have been observed in American samples [21], including the "maturity gap" between psychosocial and cognitive capacities [43], suggesting an endogenous, rather than cultural, explanation for these patterns. As youth mature into adulthood, further brain development improves communication between the prefrontal cortex and the emotional centers of the brain, such that emotion regulation and behavioral control increase, and impulsivity and sensation seeking decrease.

The adolescent years are therefore a seminal time during which youth acquire the tools, psychosocial skills, and identity that form their adult life. However, they are also a high-stakes time, during which social inputs may have outsized consequences for eventual adult beliefs. Critically, trust represents one of the key developmental tasks of this period. Beyond its long-term implications for adult orientations, trust in adolescence is closely tied to proximal outcomes such as mental health and well-being, the quality of peer and romantic relationships, and the development of social skills and competence [63, 65]. Youth who develop trusting relationships have stronger social support networks [51, 64], while mistrust has been associated with heightened loneliness [59]. Given well-established neural correlates to trust among adults [37], the neurodevelopmental changes that occur during adolescence have enormous implications for how youth trust, with consequences both for immediate psychosocial functioning and for the shaping of durable adult orientations toward others.

2 The developmental underpinning of youth trust

A number of developmental capacities are in flux during adolescence, which impact how adolescents reason and interact with their social world. Thus, it is no surprise that longitudinal research finds that social trust is least stable during the teen years, before crystalizing in the early- to mid-twenties [1], tracking the timeline for brain development. In this section, I describe how adolescent brain development impacts cognitive skills, facial processing, and social interactions with peers and parents, as well as how each of these developmental considerations underpin trust.

2.1 New cognitive skills

One of the primary developmental changes that takes place during adolescence is the advancement of cognitive skills [23]. These skills can be grouped into three domains that have implications for how adolescents conceptualize and enact trust. The first domain is abstract reasoning. During adolescence, youth become increasingly capable of abstract thought, enabling them to think in "shades of gray" rather than black-and-white terms [22]. This flexibility allows adolescents to evaluate trustworthiness in more nuanced ways, rather than relying solely on concrete rules or immediate experiences. The second domain is argumentation and evaluative judgment. Adolescents develop greater facility in constructing and evaluating arguments [50]. This cognitive shift enables them to weigh competing perspectives and evidence when deciding whether to extend trust or withhold it. The third domain of key developmental change is the strengthening of perspective-taking, or the ability to recognize that others' thoughts, intentions, and motivations may differ from one's own [34]. This "mentalizing" capacity underpins the ability to anticipate others' behavior and assess trustworthiness in increasingly complex social contexts.

These new cognitive capacities support a gradual shift from rudimentary understandings of fairness and trust, observable even in very young children [25, 27], toward more sophisticated, adult-like conceptualizations of trust and distrust during adolescence [33, 35, 77]. Longitudinal studies confirm age-related changes in both trust and distrust, reflecting adolescents' growing ability to integrate abstract reasoning, evaluative judgment, and perspective-taking into their social decision-making [71].

Brain imaging research helps explain these developmental shifts. Age-related increases in trust correspond with maturation in brain regions responsible for perspective-taking and mentalizing [28, 78], particularly when evaluating friends relative to unknown peers [75]. Similarly, distrust decisions become more calibrated as adolescents gain improved inhibitory control and conflict-monitoring capacities [28]. Taken together, this evidence suggests that adolescent cognitive development enhances the accuracy of trust and distrust judgments. Indeed, much of the brain circuitry responsible for social behaviors—including trust and distrust—develops rapidly during this period [33].

2.2 Facial processing

A particular structural and functional change in brain development that occurs during adolescence is worth highlighting as it pertains to trust. An accurate understanding of another person's mental state may hinge on interpreting their facial expression [85]. Indeed, research suggests that children can detect nuanced distinctions in facial trustworthiness [69]. However, brain imaging research suggests that the area of the brain used to process and correctly interpret facial expressions changes from adolescence to adulthood, corresponding to improvements in the accuracy of facial processing (see review in [8]). Thus, it is no surprise that a brain imaging study found several developmental changes in the neural systems activated when youth were asked to evaluate the trustworthiness of a face, relative to those activated when adults make similar judgements [49]. Importantly, meta-analytic research reports that the area of the brain used in adolescent assessments of facial expressions, the amygdala, plays a predominant role when processing negative social signals, and is thus highly implicated in distrust judgments [66].

2.3 Peer socialization and reward sensitivity

The parts of the brain implicated in social processing (i.e., the social brain), mature both structurally and functionally during adolescence [45]. As a result, adolescents are highly sensitive to social rewards [11], which increases their susceptibility to peer influence and sensitivity social to rejection in a way that directs both pro- and anti-social behavior [4]. Although one's social context spans many social interactions across levels, parents and peers are adolescents' two primary socializers [40], both of whom socialize youth trust [60].

Adolescents are neurologically inclined to find trust in peers socially rewarding, as different brain regions are engaged when evaluating trust in social versus non-social contexts. For example, fMRI research finds that adolescents use different parts of the brain when deciding to trust in a socially rewarding situation (i.e., a friend) compared to in a situation where no social rewards are present (i.e., unknown peer; [75]). Conversely, when adolescents interact with a trustworthy partner, the reward centers of the brain are activated [28]. Taken together, this research suggests that youth—who are already

neurologically primed to seek out social rewards—find trust in peers to be socially rewarding.

Given the neural salience of peer relationships and social rewards, it is no surprise that youth's trust in others is socially-motivated. For example, youth aged 12–18 were given prior social information about a partner (i.e., that they were trustworthy or not trustworthy) before playing a trust game. Younger youth who entered the task believing that their partner was trustworthy were more likely to share with this partner, and continued to rate them as trustworthy even as the partner's behavior became less trustworthy. With age, youth were better able to overcome prior social information about their partner to form their own judgement of the partner's trustworthiness based on the partner's behavior. By late adolescence, participants were able to reward trustworthy behavior from a partner they'd previously been told was untrustworthy [52]. These results suggest that younger youth are ill-equipped to make accurate trust judgements when faced with contradictory social information, due to the premium placed on social settings, an ability that improves with age.

Likewise, while early adolescents ages 9 and 12 years old did not show different patterns of trust behavior (i.e., when interacting with friends, antagonists, neutral classmates, and anonymous peers) in an allocation game, adolescents ages 15 and 18 years differentiated trust behavior according to relationship, displaying the most trust towards friends. Thus, the desire for social rewards from friends, mediated by perspective-taking skills, may drive trust differentiation during adolescence [36]. This age-related difference in trust judgements is reflected in fMRI research, which demonstrates that younger adolescents activate different brain regions implicated in social cognition than emerging adults when making trust decisions [78]. Indeed, researchers suggest that contextual factors such as social rewards override the impact of personality traits on trust behavior among adolescents [72].

Taken together, it is clear that adolescents' trust judgements are heavily influenced by developmentally-salient social rewards and peer relationships, with their ability to evaluate trustworthiness improving as they mature. Younger adolescents are particularly susceptible to prior social information, while older adolescents become more adept at forming judgments based on observed behaviors rather than preconceived notions.

2.4 Parental socialization

Developmental science suggests that parents are the basis upon which youth form their attitudes, values, and big-picture orientation toward other people [17]. Thus, it is natural that trusted adults shape how youth learn to trust others. Parents' impact on youth trust happens through two mechanisms: first, by explicitly socializing youth to hold the same values that they hold; and second, by sending indirect messages about trust through their parenting practices.

Parents socialize their children's trust-related values through explicit messages. Much of this research has been done in the context of how youth develop trust in the justice system. For example, research with justice-involved families finds that parents socialize their children's trust in the justice system generally [12] and law enforcement specifically [13]. However, parents also influence adolescents' orientation toward trust more generally as well. A longitudinal study of 11–18-year-olds found that adolescents are socialized to higher levels of trust when their parents transmit values of compassion

for others, while parents' cautions about being taken advantage of were associated with lower socialized trust [83].

Parents also transmit indirect trust-relevant lessons to their children through their behavior. Longitudinal research finds that adolescent social trust is predicted by parental emotional warmth [54] and a democratic parenting style [83]. Likewise, a secure attachment style between parent and child supports calibrated trust and social competence (see review by [74]). Conversely, age-related gains in trust behavior were attenuated in youth ages 14–25 who experienced family adversity [61], suggesting that parenting behaviors may stymie trust. For example, adverse childhood experiences (ACEs), many of which are focused on experiences in the home, are associated with mistrust and vigilance in social interactions [47, 58]. Overall, both explicit and indirect messages from parents may inculcate children to trust (or distrust) others.

3 Contemporary contextual influences on adolescent trust

Adolescent trust develops within a shifting social and historical context. Two contextual influences are particularly salient and warrant explicit attention: the increasing digital nature of social interaction, and large-scale societal stressors. While these contextual influences likely exert influence on trust dynamics regardless of age, it is critical that trust research grapple both methodologically and theoretically with the implications of modern contexts on youth trust specifically.

Increasingly, trust-relevant social interactions occur online, where digital communication can both broaden opportunities for social connection and heighten risks of misinformation or betrayal [31]. Social media platforms alter cues available for trust judgements (blurred identity signals, asynchronous communication, quantifiable social approval), which can amplify both trust and mistrust depending on platform affordances and moderation practices [41, 57, 81].

Furthermore, modern global and structural challenges may undermine adolescents' confidence in institutions and amplify mistrust in relationships. These challenges may include the COVID-19 pandemic, which disrupted interpersonal routines and institutional relationships, reshaping perceptions of health authorities, educators, and political institutions [76]; the climate crisis, which may alter institutional and interpersonal trust [73]; widening social inequality, a well-known correlate of trust [44]; and housing instability, each of which create conditions that undermine trust generally.

When these societal forces intersect with developmental changes in cognition and social motivation, they may heighten the salience of trust and mistrust as adolescents navigate not only their immediate social worlds but also their place in a broader, and often unstable, global community. Thus, any account of adolescent trust development must attend to these contemporary influences alongside normative developmental processes. Indeed, these contemporary contextual influences provide fertile ground for methodological and theoretical expansion of the adolescent trust literature.

4 Implications for adult trust

Adolescence is a critical period for the development of social trust, influenced by both cognitive and psychosocial changes as well as social interactions with peers and parents. During this time, advances in cognitive skills, such as abstract thinking and understanding others' perspectives, contribute to a more nuanced understanding of trust. However,

heightened sensitivity to social rewards can make adolescents' trust judgments particularly susceptible to peer influence. Parental influence is also pivotal, with parents shaping youth trust through explicit socialization of values and indirect messages conveyed through their behaviors. Taken together, these complex developmental factors have important theoretical and practical implications for how adult trust is formed.

4.1 Theoretical implications

Youths' trust likely shapes their ability to trust in adulthood, as the foundational attitudes and behaviors developed during this formative period have lasting impacts on attitudes and behaviors later in life [30, 79]. During the transition to adulthood, individuals must navigate increasingly complex social dynamics [82]. These early experiences interact with brain development to create a framework through which young people evaluate future relationships and societal interactions [70].

Understanding adult trust, particularly through established models like the ABI (Ability, Benevolence, Integrity; see Mayer et al. [55]) model, requires acknowledgment of these adolescent legacies; it may be that trust is not merely a function of situational factors in adulthood but is influenced by earlier experiences. To integrate developmental perspectives into existing trust theory, researchers should consider how early experiences calibrate expectations about trustee ability, benevolence, and integrity over time, as well as how these calibrated expectations interact with cognitive maturation to produce adult patterns of generalized and institutional trust. Specifically, integrating developmental theories such as social learning theory and attachment theory with the ABI model provides a richer account of how trust evolves. Social learning theory helps explain how observational learning during adolescence shapes expectations about others' ability, benevolence, and integrity. Attachment Theory illuminates how early relational patterns create templates for evaluating trustworthiness that persist into adulthood. Erikson's concept of basic trust provides a lens for understanding how adolescent experiences either reinforce or challenge foundational trust orientations established in childhood.

4.2 Practical implications

A holistic conceptual understanding of youth trust also has important practical implications for adult life. At the cusp of adulthood, youth rapidly begin to gain societal rights and responsibilities. In many places, they are able to make medical decisions for the first time during the teen years. Yet, a comprehensive understanding of how youth develop trust in healthcare providers is lacking [39]. Likewise, the global median age at which youth are enfranchised is 18 [2], with calls in many developed nations to lower the voting age to 16 [24]. However, evidence from a study of political trust in five countries across over 50 years suggests that political trust is highly unstable during the transition from adolescence to adulthood [19]. A developmental lens may illuminate the mechanisms through which political trust changes and crystalizes at the very time when youth are entering the ranks of the voting citizenry. Lastly, adolescents are more likely than any other age group to have interactions with the justice system, yet our understanding of how they develop trust in legal authorities is still incomplete, despite its known association with continued law-breaking behavior [12] and evidence of decreases in trust in legal authorities over the last fifty years [29]. Gaining a comprehensive,

developmentally-informed understanding of adolescent trust across various systems and domains is crucial for supporting youth as they navigate significant societal transitions into adulthood.

5 Future directions in trust research

5.1 Methodological approaches

Methodologically, trust research can most effectively advance toward a more complete understanding of trust by recognizing that the adolescent years are a developmentally distinct point in the lifespan. Researchers should carefully consider sampling strategies; youth brain development continues through the mid-20s, yet much trust research considers anyone above the age of 18 an adult participant (undergraduate convenience samples come to mind). Combining youth and adult participants may obscure important findings or lead to misinterpretations of age-related differences. For example, one manifestation of youths' heightened sensitivity to social rewards is their increased likelihood to engage in risk-taking behaviors, which peaks in the late teens and early 20s. This propensity is not necessarily due to greater trust in their peers, but rather a desire for social approval or reward [46]. This distinction is particularly important for trust researchers, as behaviors driven by social motivation during youth may be mistakenly interpreted as indicators of trust.

Additionally, much of the current research specific to adolescent trust uses the Trust Game [6], a methodological approach with well-documented limitations [3]. The Trust Game fails to capture critical social mechanisms relevant to youth, such as peer approval, and is not reflective of real-world social trust, which may be neurally-rewarding to adolescents. Instead, it presents a cognitively complex economic problem within a social vacuum. To address these shortcomings, future work could integrate experimental paradigms that embed trust decisions in developmentally salient contexts. This could include modified trust games that include digital reputational feedback, opportunities for reciprocity, or peer evaluation. Experience-sampling methods and ecological momentary assessment could also provide insight into how trust operates in adolescents' daily lives, while longitudinal survey approaches can track trajectories of trust alongside related outcomes such as well-being and social competence. Finally, neuroimaging tasks that incorporate realistic social interactions, particularly as mediated by social media, may help balance the strengths of controlled paradigms with greater ecological validity. Together, these approaches would expand the measurement toolkit beyond the traditional Trust Game, offering a more nuanced and developmentally sensitive account of adolescent trust.

5.2 Conceptual approaches

Future conceptual approaches to trust research could also benefit from adopting a developmental lens. Current theories of trust do not explicitly account for the qualitatively different neurology, capacities, and influences that adolescents experience relative to young children or to adults [67]. This is true both when considering youth trust explicitly, but also in considering adult trust, which, as described above, is necessarily formed by experiences and values developed earlier in life.

There is enormous opportunity for expanding theories of trust research by incorporating what is known about youth development. As just one example, there is currently a

dearth of research on how one of the most salient aspects of adolescent brain development—psychosocial maturity—is associated with trust. Psychosocial maturity is associated with impulse control, decreased sensation-seeking (i.e., the propensity to pursue new and thrilling experiences), and the ability to consider future consequences, and is the focus of a large share of developmental psychological research (Icenogle 2019). Because psychosocial maturity does not reach adult levels until the mid-twenties, youth are generally more impulsive, more sensation-seeking, and less able to consider consequences than adults. These developmentally-normative differences mean that youth view and interact with the social world differently than adults. To date, however, very limited research provides oblique empirical clues as to how youth trust may be filtered through their psychosocial capacities.

Some of the key components of psychosocial maturity have been studied only among adults. First, there is some evidence that increased impulsivity is associated with increased trust in young adults (i.e., college undergraduates; see Ibañez et al. [42]). Second, among adults, trust and sensation seeking, or the propensity to pursue new and thrilling experiences, are positively correlated [14, 84]. However, no research appears to have covered how, whether, and in what circumstances these constructs are related in adolescence, a time when sensation-seeking reaches its zenith. Lastly, hopeful future expectations were positively correlated with trust in a sample of adolescents, and revealed four distinct trajectories three years, suggesting that future expectations and trust move together through development [10], however this research operationalized future expectations as aspirations rather than the ability to consider consequences.

Given that the characteristics comprising psychosocial immaturity, a hallmark of adolescence, may be associated with trust among adults, there is an opportunity for developmentally-informed research on how youths' relative psychosocial immaturity may be associated with trust. Additionally, because individual youth display different trajectories of psychosocial development [56], understanding how psychosocial maturity is associated with trust in adolescence may help contextualize trust in adulthood.

Additionally, there is opportunity for future research to explicitly integrate classical developmental theories with trust theory to understand how youth trust develops. For example, Social Learning Theory predicts that adolescents who observe trusting behavior from role models should be more likely to exhibit trust themselves, but the neural sensitivity to social rewards during adolescence may amplify or moderate this effect in ways not yet empirically examined. Similarly, Attachment Theory suggests that secure attachment should promote calibrated trust, yet how this process unfolds amid the neurodevelopmental changes of adolescence remains underspecified. Integrating these established developmental theories with contemporary neuroscience and trust literature would strengthen both theoretical precision and empirical prediction in adolescent trust research.

6 Conclusion

Adolescence represents a critical period for the development of trust, shaped by the interplay of cognitive, psychosocial, and social factors. The distinct developmental changes that occur during these years—from advancements in cognitive skills to heightened sensitivity to social rewards—profoundly influence how adolescents perceive and navigate trust in various domains. Yet, the trust literature has not undertaken a

concerted effort to understand trust among young people. Indeed, in all journals published in the Journal of Trust Research from 2011 to 2024, the words "child" "children" "adolescent" "adolescence" "youth" or "young" appear in only one article, in which parent-child relationships are one of several trust relationships addressed. I advocate for adopting a developmental lens in trust research, both to recognize the unique challenges and opportunities that characterize adolescent trust and also to understand how youth trust shapes adult trust. By integrating insights from developmental science, trust researchers can advance the literature and build the theoretical foundation for fostering more effective strategies for building and maintaining trust during this formative stage of life.

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